maturity. Fixed income investments, or bonds, typically provide income and have low historical volatility.

Asset Allocation—Flaws in Mean-Variance Optimization?

While Markowitz's pioneering research in the 1950s highlighted the benefits of diversifying across multiple asset classes, there has been substantial debate over the ensuing decades on the best way to achieve this. Since investors can generate varying rates of return and assume alternative levels of risk by combining quantities of asset classes in different ways, this debate has focused on the development of theoretical models that have provided insight into an array of weighting methodologies.

These methodologies have offered competing approaches to optimizing portfolio performance over time, but they are all based on attempts to simultaneously maximize return while assuming the least amount of risk, also referred to as *mean-variance optimization*.

Despite the diversity and sophistication of these models, there are two criticisms common to all of them: (1) the length of the sample periods tested, which are short, and (2) the estimates of return and volatility used to calculate the allocation rules, which can change in the future. Fundamentally, these models are derived from uniformly small samples whose results often do not hold up when implemented over long, out-of-sample, time frames, and their return and volatility estimates, also when applied out-of-sample, tend to be noisy and prone to error over time, which often results in the models' diverging, sometimes radically, from predicted outcomes.

As it turns out, this problem is not unique to financial markets.

As Daniel Kahneman points out in his 2011 book, *Thinking Fast and Slow*: